MECH-3212 Thermodynamics **MATLAB Assignment**
Due by the end of class, Monday, 13 November 2023

Heat Exchanger Design

In this assignment you will use MATLAB to investigate the effect of inlet temperature on the required mass flow rate in a tube-in-tube heat exchanger.

A tube-in-tube heat exchanger is to cool ethylene glycol, c_P = 2.56 kJ / (kg·°C), flowing at a rate of 3.2 kg/s from an inlet temperature of 80°C to an outlet temperature of 40°C;. The other fluid flowing in the heat exchanger is liquid water, c_P = 4.18 kJ / (kg·°C). Investigate the effect of the inlet temperature of cooling water on the mass flow rate of the water. Let the inlet water temperature vary from 10°C to 30°C in intervals of 2°C, and assume the exit temperature of the water remains constant at 70°C.

- You will solve this problem by 2 methods:
  1) $\Delta h = c_P \Delta T$ with a constant specific heat
  2) For compressed liquid water, input the h_R values from Table A-4 from 10°C to 30°C and interpolate to find the input saturated liquid enthalpy value for the given temperature range.
- You will plot the results of both methods on one graph, with inlet temperature on the x-axis, and mass flow rate of water on the y-axis.
- The program will output a data ledger, with the following information.
  - Column 1 - inlet temperature
  - Column 2 - mass flow rate based on constant specific heat
  - Column 3 - mass flow rate based on saturated liquid enthalpy

To be eligible for full credit, you will submit the following items. The section below on marking scheme has additional information.

- A cover sheet with your names, e-mail address, and signatures
- A copy of your commented MATLAB code - define all variables, explain what each section of the code does
- Printouts of your graphs and data ledgers
- One paragraph commenting on your results, word-processed, not hand-written.

Marking Scheme

3 marks – The program should be well commented (%), and have a logical design and flow.

3 marks – The program should perform the calculations properly and return correct answers. The output information should appear in a data ledger as noted above. The output plot should include a title, axis labels with units, major and minor subdivisions on the axes. Data for both methods should be plotted with unique symbols and separate lines.

3 marks – You will include a paragraph that comments on your results. Include the range of calculated mass flow rates; any variance (or lack of variance) between the two methods; sources of error.

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Submission Guidelines and Late Penalties

- The "end-of-class" deadline on Monday, November 13 is firm. If you cannot attend class that day, you must submit your assignment early to receive full marks.
- For assignments received after 12:50 pm and up to 4:30 pm on Monday, November 13, the earned mark will be reduced by 25%. After 4:30 pm on November 13, Mini Projects will not be accepted.
- You must submit your paper in person to Dr. Bowers or any of the GA's in class, during office hours, or in Dr. Bowers's office when he is there. No assignments placed in mailboxes or slipped under doors will be accepted.

Academic Dishonesty

- Do not delete your files upon completion of the project.
- In the event that your code and/or results are deemed to be similar to those of another student, you will be required to submit your MATLAB code files to Mohammed Ismail for further inspection.
- In instances of plagiarism, a mark of zero will be awarded to both the originator(s) of the code and the person(s) who copied/borrowed all or portions of the code. All instances of academic dishonesty will be submitted to the Academic Integrity Office.